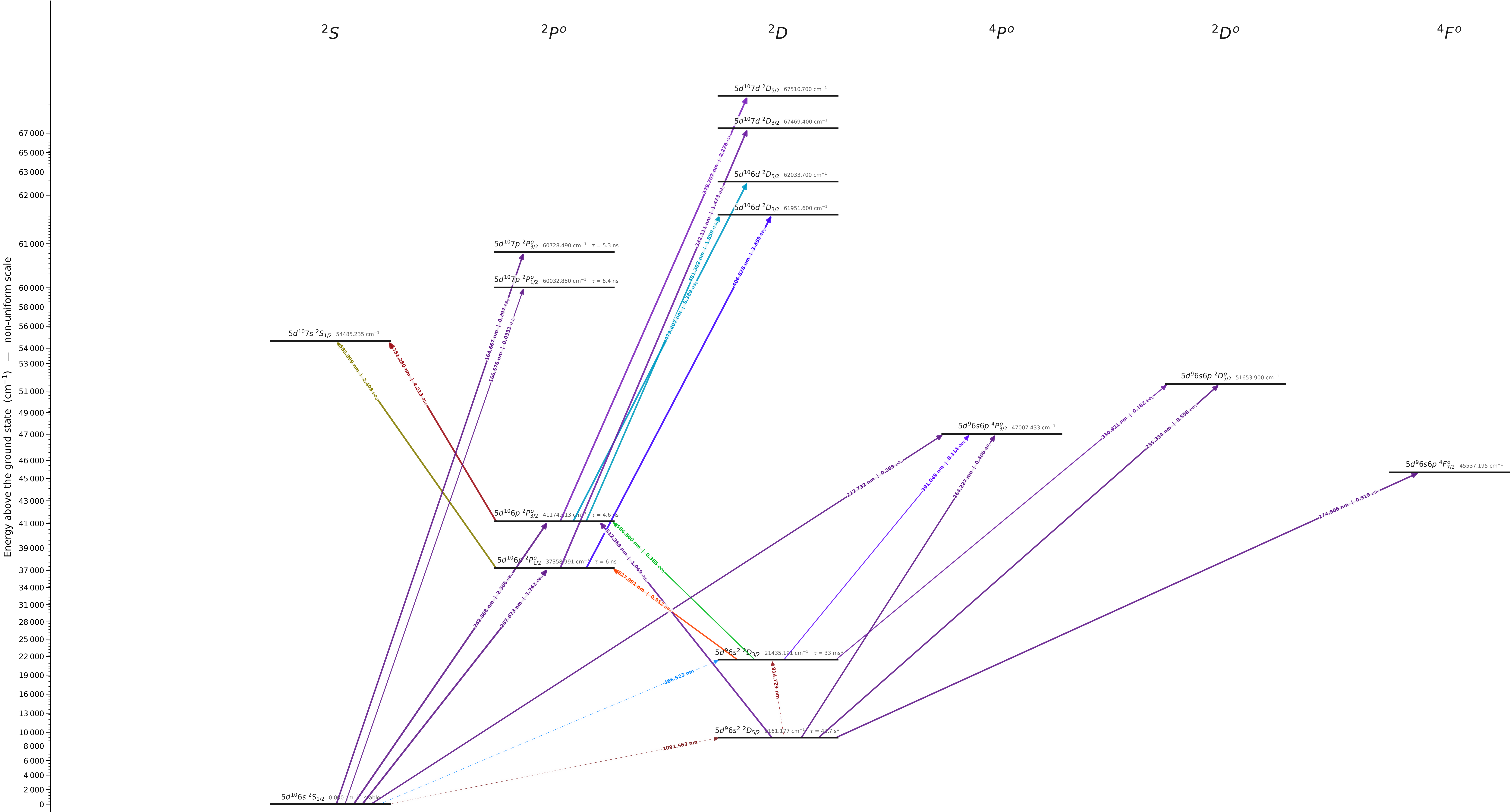


Au (Au I). Strongest classified transitions below 2 μm (NIST): 15 levels, 23 lines.

Au⁺ ionisation limit 74409.11 cm⁻¹ = 9.22555 eV (134.392 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle V \| er \| J'J'' \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U e U' \rangle$ (ea0)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$5d^96s^2\ ^2D_{5/2} - 5d^96s^2\ ^2D_{3/2}$	814.7294	814.5054	367.9657	-	2.990e+01	4.823e-06	98.7	main decay of $2D3/2$
$5d^96s\ ^2S_{1/2} - 5d^96s^2\ ^2D_{5/2}$	1091.5628	1091.2639	274.6452	-	2.290e-02	3.642e-09	100	E2 clock transition proposed for a Au lattice clock

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

¹⁹⁷Au hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$5d^{10}6s\ ^2S_{1/2}$	3049.660092(7)	-	F = 1: -3812.07 F = 2: +2287.24
$5d^96s^2\ ^2S_{5/2}$	80.236(3)	-1049.781(11)	F=1: -1156.09 F=2: -155.79 F=3: +557.32 F=4: +38.44
$5d^96s^2\ ^2D_{3/2}$	199.84250(20)	-911.0766(5)	F=0: -1888.26 F=1: -777.34 F=2: +533.43 F=3: +221.88
$5d^{10}6p\ ^2P^o_{1/2}$	321.7(12)	-	F = 1: -402.12 F = 2: +241.28
$5d^{10}6p\ ^2P^o_{3/2}$	14.0(5)	327.6(16)	F=0: +357.00 F=1: +43.40 F=2: -256.20 F=3: +113.40

Measured A, B constants; sources in the data file.